AskMyDocs: LLM-Powered Document Summarization & Semantic Information Retrieval

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Abstract
AskMyDocs is a Retrieval-Augmented Generation (RAG) web application that allows users to upload PDF documents and query them using natural language. The system combines semantic search via FAISS vector stores with large language models (LLMs) hosted on Groq to produce accurate, grounded answers with source citations. The application supports both text-based and scanned PDF documents through an integrated OCR pipeline using PyMuPDF and Tesseract.

Chapter 1: Introduction to Retrieval-Augmented Generation

Retrieval-Augmented Generation (RAG) is a hybrid AI approach that combines the strengths of information retrieval systems with generative language models. Unlike pure generative models, which rely solely on learned parametric knowledge, RAG systems augment generation with dynamically retrieved context from external knowledge bases.

Key Benefits of RAG:
1. Grounded responses — answers are based on actual documents, reducing hallucination.
2. Updatable knowledge — documents can be swapped without retraining the model.
3. Source attribution — users can verify the provenance of each answer.
4. Domain adaptability — works across legal, medical, technical, and academic domains.

The RAG pipeline in AskMyDocs consists of nine stages: document upload, text extraction or OCR, preprocessing and chunking, embedding generation, FAISS vector store indexing, query embedding, semantic retrieval, LLM generation, and structured answer output.

Chapter 2: System Architecture

AskMyDocs follows a modular five-layer architecture:

Layer 1 — Presentation: A Streamlit web interface with tabs for Chat, Summary, Documents, and Logs. Users upload PDFs, configure retrieval parameters, and interact with the system through a clean, responsive UI.

Layer 2 — Processing: PDFs are loaded using PyPDF and PDFPlumber. Pages with fewer than 50 extractable characters are automatically routed to the OCR module, which uses PyMuPDF to render pages at 300 DPI before passing them to Tesseract.

Layer 3 — Embedding and Retrieval: The all-mpnet-base-v2 model from Hugging Face generates 768-dimensional embeddings. These are stored in a FAISS flat index persisted to disk. At query time, the top-k most similar chunks are retrieved using cosine similarity or Maximum Marginal Relevance (MMR).

Layer 4 — Generation: LangChain orchestrates a ConversationalRetrievalChain that condenses follow-up questions into standalone queries, retrieves context, and prompts the selected LLM. Groq's Llama 3.x models serve as the primary provider.

Layer 5 — Output: Each answer is displayed with collapsible source expanders showing the filename, page number, and a short text snippet for verification.

Chapter 3: Technology Stack

The AskMyDocs system is built on the following technologies:

- Python 3.11+: Core programming language.
- Streamlit 1.44: Web framework for the interactive user interface.
- LangChain 0.3.x: RAG orchestration, chain management, and memory.
- FAISS-CPU 1.9: High-performance vector similarity search.
- Sentence Transformers (all-mpnet-base-v2): Local embedding model, no API key required.
- Groq API: Cloud LLM inference using Llama 3.3 70B Versatile and Llama 3.1 8B Instant.
- PyPDF and PDFPlumber: Text extraction from text-layer PDFs.
- PyMuPDF (fitz): Page rendering for the OCR pipeline.
- pytesseract + Tesseract: OCR engine for scanned documents.
- python-dotenv: Secure loading of API keys from .env files.

All LLM inference runs through cloud APIs, making the application suitable for standard laptops without GPUs. Embedding generation runs locally via CPU, cached using Streamlit's st.cache_resource decorator.

Chapter 4: Evaluation and Future Work

Evaluation Methodology:
The system is evaluated across four dimensions:
1. Retrieval Accuracy (Hit@k): Whether the expected source page appears in the top-k results.
2. Summary Quality: Human evaluation of short, detailed, and bullet-point summaries.
3. OCR Accuracy: Character Error Rate (CER) computed against ground-truth transcriptions.
4. Response Latency: Time from query submission to full response across different models.

Preliminary Results:
- Llama 3.3 70B Versatile achieves the highest answer quality with average latency of 2.3s.
- Llama 3.1 8B Instant is 3x faster with slightly lower verbosity.
- FAISS retrieval with all-mpnet-base-v2 achieves Hit@6 of 0.91 on the test set.

Future Scope:
- Support for DOCX, TXT, and Excel document formats.

- Multilingual document queries using multilingual embedding models.
- Voice interaction through speech-to-text (Whisper API).
- Cloud deployment on AWS/GCP with Streamlit Community Cloud.
- Fine-tuned retrieval using domain-specific embedding models.